CDS6314 DATA MINING
Trimester 2530

PROJECT

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Abstract

Banks are having more and more trouble with customer attrition, which is when they lose existing customers. Keeping existing customers is much cheaper than getting new ones. This project makes a tool that helps bankers to find credit card customers who are likely to leave. Using a structured Kaggle dataset containing demographic and transactional behavior, we implemented a two-pronged data mining pipeline, relationship discovery via association rule mining (FP-Growth) and model classification prediction.

The models we evaluated for predicting attrition are Logistic Regression, Support Vector Machine (SVM), Random Forest (RF), and Gaussian Naive Bayes. The Random Forest model demonstrated the strongest overall performance with a ROC AUC of 0.9852. These models, alongside interpretable churn patterns derived from association rules, were integrated into an interactive Streamlit application. This tool allows non-technical users to visualize portfolio risk, explore behavioral profiles, and simulate individual customer churn probability to guide practical retention strategies.

Introduction

Background

Banks are negatively affected by customer attrition or churn in their portfolios, retaining customers is typically cheaper than acquiring new customers and small changes in customer retention can affect banks profits. This project analyzes a credit-card customer dataset to discover patterns and produce an operational alerting tool that helps business users identify at-risk customers.

Dataset

For this project, we use a structured credit card customer dataset from kaggle called “Predicting Credit Card Customer Segmentation”. This dataset size is 10127 rows and

21 attributes, contains demographic information about the users, account characteristics, and transaction behaviour. With so much customer information from a consumer's credit card portfolio, we can use it to analyze the key factors that are influencing customer attrition. The data can be analysed to better predict customer attrition. If further built upon this project could assist banks to detect early signs or dissatisfaction or disengagement with the credit card.

Dataset link: [Predicting Credit Card Customer Segmentation]

Motivation

Understanding customer attrition patterns is very important and practical retention endeavors require tools. These tools must be usable by non technical workers. This project hopes to provide such tools, whereby analytical results can be translated into an interactive decision support tool for bankers. The application allows users to view customer attrition rate, get insight into each customer's relationship, and obtain churn prediction for individual customers based on imputed attributes. This motivates the inclusion of both exploratory analysis and predictive modelling in the project, as well as the development of an application layer to bridge analysis and practical usage.

Objectives

- Use Exploratory Data Analysis to understand customer demographic, behavioural patterns and the difference between existing and attrited customers.
- Use association rule mining techniques to relate common combinations of customer attributes to attrition.
- Develop supervised classification models that can predict customer attrition.
- Develop an interactive application that uses the developed models to allow users to input customer information and evaluate attrition risk.

Related Works

This project is threading on a heavily explored field thus much research has been done in this field, we reviewed some of these works to get ideas and guide us to completion. Among the works is a study by Al-Najjer et al. (2022) which investigated credit card customer churn predicting on a dataset consistent with the publicly available Kaggle BankChurners dataset. The authors evaluated multiple machine learning models such as decision trees, Bayesian networks and neural networks using different feature selection strategies. They found that tree based models have the best performance and that transaction related behavioural features were the best predictors. Similarly, Tran (2024) applied supervised machine learning techniques directly to the BankChurners dataset and demonstrated that customer behaviour and usage patterns play a significant role in distinguishing attrited customers from existing ones. Calp (2025) also utilised the same Kaggle credit-card dataset, focusing on feature selection and predictive modelling to improve churn prediction accuracy.

Most of the studies reviewed primarily emphasised predictive performance and model comparisons as such they can be classed under classification alone. In our project we attempt to extend prior work by combining supervised churn prediction with association rule mining to provide interpretable insights into common churn patterns. We also include an interactive application to demonstrate how predictive models and explanatory patterns can be integrated into decision support machines.

Methodology

In this section we discuss the pipeline we used to complete the project. The project adheres to the standard data mining pipeline which include exploratory data analysis, data preprocessing, data selecting and knowledge discovery followed by modelling.

Overall Framework

The project implements a two pronged pipeline which is relationship discovery using association rule mining and supervised classification. Then a Streamlit application integrates the outputs from both pipelines, the application does not recompute models or mining it uses precomputed artifacts saved in a folder. The application which is where the user, in this case a banker, interacts with the pipelines. It provides a portfolio level overview of customer risk, exploration of common customer behaviour patterns and an individual customer churn prediction or a simulation via user given attributes. Below is the general pipeline flowchart.

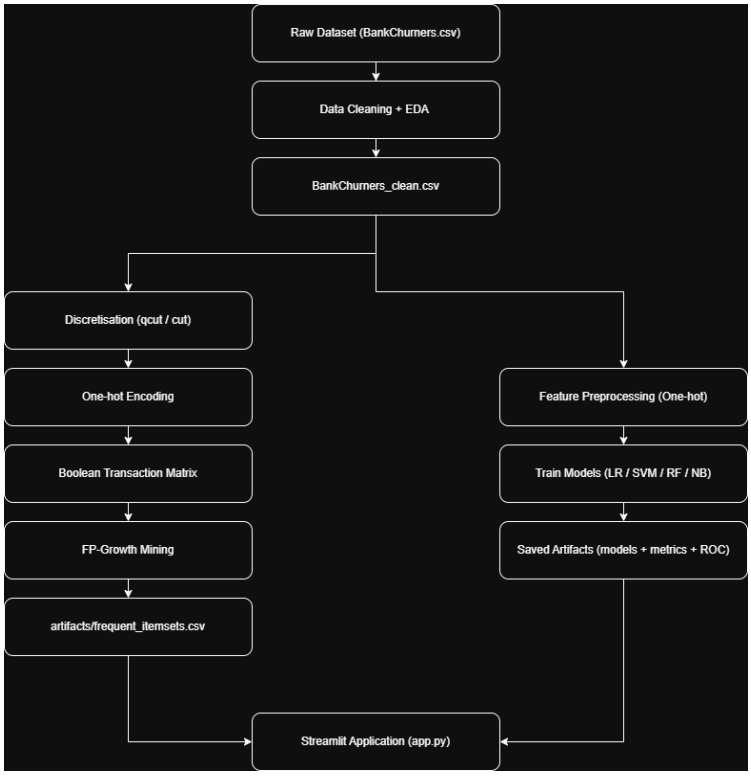


Figure 1 pipeline flowchart

Dataset

Demographic Features:

- Customer_Age: Age of customer
- Gender: Gender of customer
- Dependent_Count: Number of customer's dependent
- Education_Level: Education level of customer
- Marital_Status: Marital status of customer
- Income Category: Income category of customer

Bank & Client Relationship Features:

- CLIENTNUM: Identify client
- Card_Category: Type of card held by customer
- Months_on_book: How many months the customer has been on the books
- Total_Relationship_Count: Total of relationships customer has with the card provider
- Months_Inactive_12_mon: Number of months that customer has been inactive within twelve months
- Contacts_Count_12_mon: Number of contacts that customer has had within twelve months

Behavioral & Credit Features:

- Credit_Limit: Credit limit of customer
- Total_Revolving_Bal: Total revolving balance of customer
- Avg_Open_To_Buy: Average open to buy ratio of customer
- Avg_Utilization_Ratio: Average utilization ratio of customer
- Total_Trans_Amt: Total transaction amount
- Total+Trans_Ct: Total transaction count
- Total_Amt_Chng_Q4_Q1: Total amount change from quarter 4 to quarter 1
- Total_Ct_Chng_Q4_Q1: Total count changed from quarter 4 to quarter 1

Target Variable: Attrition_Flag, 1 = churned, 2 = not churned

Data Preprocessing & EDA

Exploratory Data Analysis is done to understand the distribution of features and potentially identify relationships between variables as well as give easily interpretable visuals. Numerical attributes are studied via histograms and density estimates, categorical variables are studied via frequency bar charts and cross tabulation. A summary table of descriptive statistics are produced and displayed in an interactive table in the app so non-technical users can inspect the distributions and identify discrepancies.

Feature Engineering and Preprocessing

Due to the nature of our project two parallel preparations are needed depending on the downstream use. These are implemented explicitly and persisted so runtime inference and reporting use the exact transforms applied during training and pattern discovery.

Association Rule Mining

For association rule mining selected continuous variables are discretised into bins. The backend script applies these transformation:

- Avg_Utilization_Ratio to Utilization_Level via qcut where = Low/Medium/High
- Total_Trans_amt to Transaction_Amount_Level via qcut where = Low/Medium/High
- Total_Trans_Ct to Transaction_Count_Level via qcut where = Low/Medium/High
- Months_Inactive_12_mon to Inactivity_Level via cut where = Few/Moderate/Many
- Customer_Age to Age_Group via cut where = Young/Middle/Old

We use this discretisation strategy for roughly balanced category sized and fixed thresholds in meaningful domains.

After discretisation the attributes are one hot encoded. The backend drops the Attrition_Flag dummy columns from the encoding to avoid leakage and makes the resultant matrix to boolean to generate a transactional basket which will be used in FP-Growth.

Supervised Classification

For supervised classification we preserve numerical features in their continuous form. These include transaction and account related attributes such as Total_Trans_Amt, Total_Trans_Ct,

Total_Revolving_Bal, Months_on_book and other behavioural measures. We do not discretise these variables because retaining continuous values allows the models to capture more fine-grained patterns and relationships in customer behaviour.

We convert categorical variables into dummy variables using one-hot encoding. This transformation creates a fixed training matrix where each column represents a specific category level, ensuring consistency between training and inference.

Numerical features that require normalisation are scaled using a fitted scaler object. We fit the scaler during model training and reuse the same scaler during inference so that all incoming data is transformed using the same scale as the training data.

We persist the preprocessing configuration to disk, which includes the list of numerical columns and categorical columns, the full ordered set of feature columns after encoding, and the fitted scaler. This specification is saved as artifacts/preprocess_columns.json and is loaded by the Streamlit application to ensure that preprocessing performed in preprocess_X_raw and compute_portfolio_risk remains aligned with the model training pipeline.

Association rule Mining backend

After the boolean transactional matrix is made it is mined with mlxtend's fpgrowth to extract frequent itemsets. We use configurable parameters to allow for flexibility in discovering patterns. Upon generation statistics such as support, number of customers that match the condition, and churn rate for each itemset is calculated. Items that do not meet the minimum customer threshold are filtered out to make sure the rules are statistically relevant.

The resulting itemsets are saved to artifacts/frequent_itemsets.csv, with the configuration used during the mining process, this helps our app not recompute mining operations during application runtime while still allowing us to display the behavioural patterns.

Model training and evaluation

For the classification part of our project we train four supervised classification models Logistic Regression, Support Vector Machine, Random Forest and Gaussian Naive Bayes, we save the models and their individual preprocessing is saved as a joblib file. This is done to compare

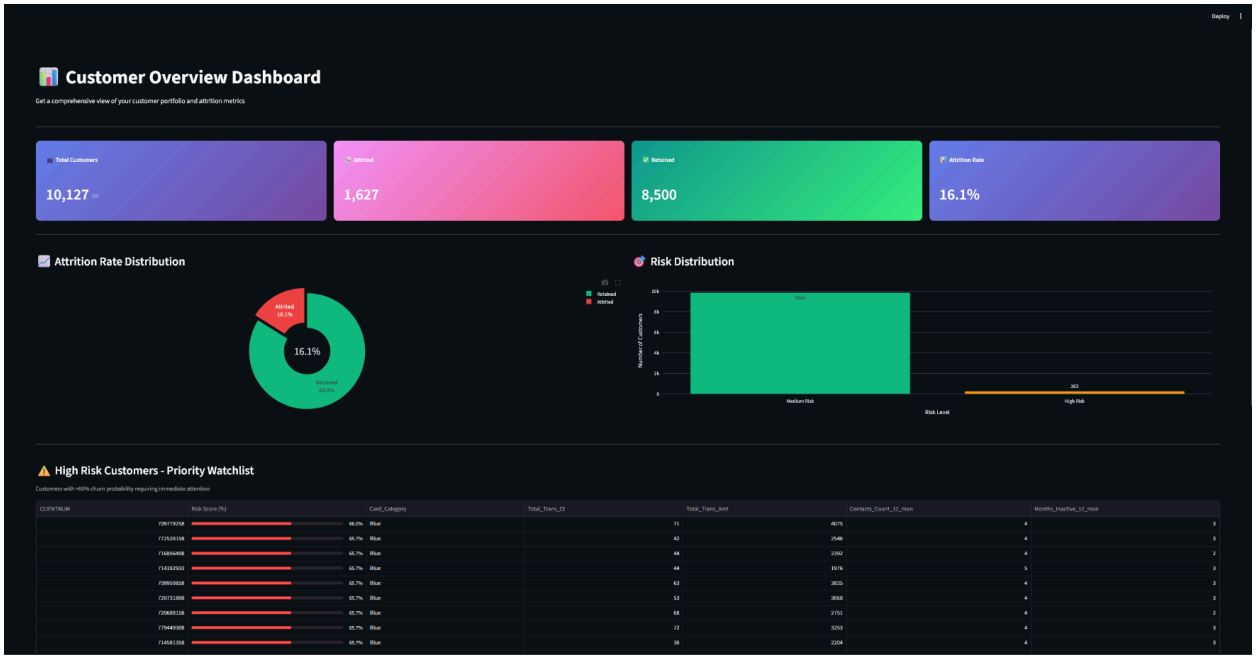
predictive performance of different models and allow the user to assist their decision making by showing multiple predictive outputs. In this stage we prepare the dataset by defining the input features and the binary target variable which is Attrition_Flag. We attempt to preserve class distribution by using stratified train test splits for the training and testing splits. To maintain consistent feature representation categorical variables are one hot encoded and numerical features are scaled via fitted scaler.

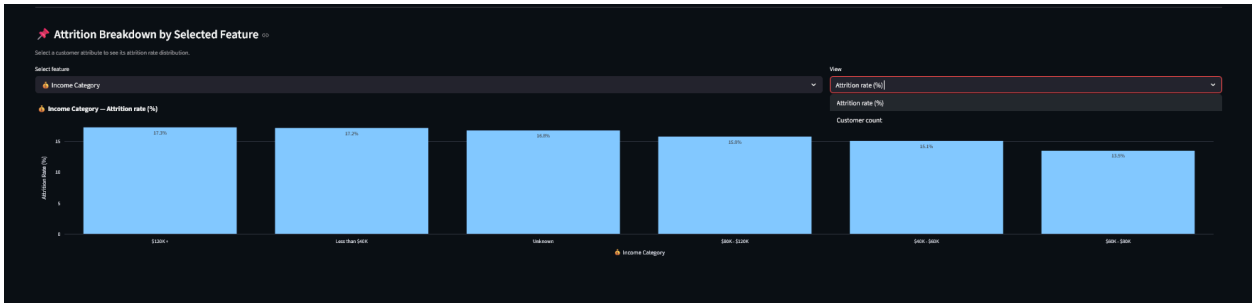
Here we discuss the model parameters, for Logistic Regression it is trained with max_iter=5000 to allow for convergence on the one hot encoded and scaled feature matrix and class_weight_balanced since attrited customers are a minority this reduces bias toward the major class. The SVM model evaluates multiple kernel types linear,RBF and polynomial using cross validation before selecting the final configuration; this is done because our dataset contains both linear and nonlinear relationships and this lets us select the best kernel that generalises best. The Random Forest model is trained with n_estimator =300 which helps reduce variance in predictions and stabilises estimates of feature importance. Gaussian Naive Bayes is used as a probabilistic baseline classifier.

After training each model is evaluated using the standard classification metrics such as accuracy,precision, recall, F1 score and ROC AUC. Confusion matrices and classification reports are generated to analyse model behaviour. After training, each model together with its associated preprocessing components is saved as a joblib file in the models directory as stated above. Evaluation results and performance metrics are saved as JSON artifacts in the artifacts directory. Additionally, ROC curve data for all models are consolidated into a single JSON file to support comparative visualisation within the Streamlit application, because we saved the trained models and evaluations the application can load and compare models without retraining at runtime.

Application

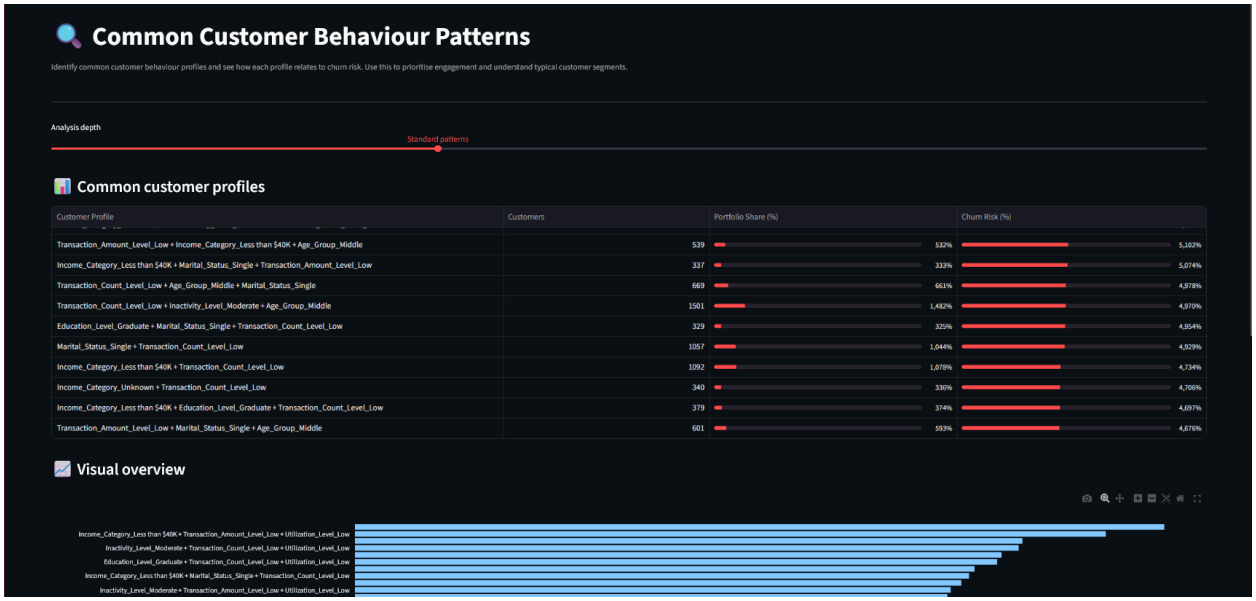
PAGE 1





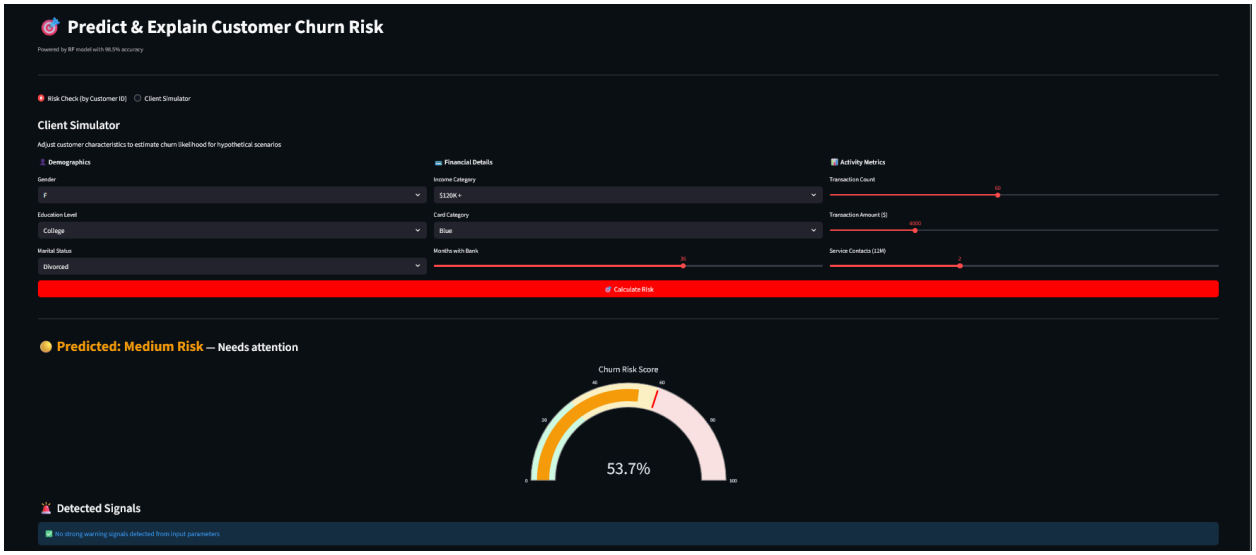
The streamlit app is where the banker will interact with the most, this is our landing page when the [app.py](app.py) file is run. At the top of the page there is a high level summary of the data set with a pie chart for the target attribute as well as risk distribution. Below that the banker can identify high risk individuals via their CLIENTNUM and this information is displayed in an interactable table. At the bottom of the page the banker can generate bar charts via a selection box, the charts can be focused on the number of customers in each category or the attrition rate.

PAGE 2



This page focuses on churn pattern exploration which is where the association rule mining comes. Instead of recomputing the mining process, this page loads precomputed frequent itemsets generated by the backend script, allowing faster interaction while maintaining consistency with the analysis pipeline. At the top of the page, the banker can select the analysis depth using a slider, which adjusts parameters such as maximum itemset length and minimum support to control the complexity and coverage of the displayed patterns. The main section presents a table of common customer behaviour profiles, where each row represents a combination of customer attributes along with the number of customers matching the profile, the portfolio share, and the associated churn risk. These patterns help the banker understand which combinations of behaviours or demographics are more frequently associated with attrition. Below that a visual bar chart for understanding at a glance is provided, it shows churn risk across different customer profiles.

PAGE 3



On page 3 we allow the banker to directly interact with the models where the bankers can input customer attributes manually and generate a predicted churn probability. The app displays the predicted risk level using a visual gauge with risk classification, allowing for fast interpretations. This page is the primary decision support interface allowing bankers to simulate clients and understand potential risk factors.

PAGE 4

	CLIENTNUM	Attrition_Flag	Customer_Age	Gender	Dependent_count	Education_Level	Marital_Status	Income_Category	Card_Category	Months_on_book	Total_Relationship_Count	Months_Inactive_12_mon	Contacts_Count_12_mon	Credit_Limit	Total_Revolving_Bal	Avg_Open_To_Buy	Total_Amt_Chng_Q4_Q1	Total_Trans_Amt	Total_Trans_Ct	Total_Ct_Chng_Q4_Q1	Avg_Utilization_Ratio	target
0	768805383	Existing Customer	45	M	3	High School	Married	$60K - $80K	Blue	39	5	1	3	12691	777	11914	1.335	1144	42	1.625	0.061	0
1	818770008	Existing Customer	49	F	5	Graduate	Single	Less than $40K	Blue	44	6	1	2	8256	864	7392	1.541	1291	33	3.714	0.105	0
2	713982108	Existing Customer	51	M	3	Graduate	Married	$80K - $120K	Blue	36	4	1	0	3418	0	3418	2.594	1887	20	2.333	0	0
3	769911858	Existing Customer	40	F	4	High School	Unknown	Less than $40K	Blue	34	3	4	1	3313	2517	796	1.405	1171	20	2.333	0.76	0
4	709106358	Existing Customer	40	M	3	Uneducated	Married	$60K - $80K	Blue	21	5	1	0	4716	0	4716	2.175	816	28	2.5	0	0
5	713061558	Existing Customer	44	M	2	Graduate	Married	$40K - $60K	Blue	36	3	1	2	4010	1247	2763	1.376	1088	24	0.846	0.311	0
6	810184308	Existing Customer	51	M	4	Unknown	Married	$120K +	Gold	46	6	1	3	34516	2264	32252	1.975	1330	31	0.722	0.066	0
7	818906208	Existing Customer	32	M	0	High School	Unknown	$60K - $80K	Silver	27	2	2	2	29081	1396	27685	2.204	1538	36	0.714	0.048	0
8	710930508	Existing Customer	37	M	3	Uneducated	Single	$60K - $80K	Blue	36	5	2	0	22352	2517	19835	3.355	1350	24	1.182	0.113	0
9	719661558	Existing Customer	48	M	2	Graduate	Single	$80K - $120K	Blue	36	6	3	3	11656	1677	9979	1.524	1441	32	0.882	0.144	0

The 4th page provide insights into the models metrics and dataset. This section allows the banker or analyst to compare model performance using evaluation metrics such as accuracy, precision, recall, F1 score, and ROC AUC displayed in an interactive table. Users can select different models to view and compare their ROC curves, providing a visual understanding of how each classifier performs across different thresholds. The page also includes a preview of the dataset for auditing purposes and offers options to export the scored dataset and model performance metrics.

Evaluation

To evaluate our models we compare the performance of the trained classification models using accuracy precision recall F1 score and ROC AUC, although we mainly focus on ROC AUC as it reflects how well each model separates attrited customers from existing customers. Relying on accuracy alone would not be enough as the dataset has class imbalance which makes accuracy a less reliable indicator or performance.

During evaluation we analyse the classification reports and confusion matrices to understand how each model behaves beyond a single score, particularly how well the models identify attrited customers since they represent the minority class, this helps us observe the balance between correctly detecting churn cases and avoiding excessive false positives which could lead to unnecessary intervention actions. We also compare ROC curves between models to visualise how performance changes across different thresholds rather than relying on one fixed

prediction cutoff, allowing us to see which model maintains stronger separation capability overall.

Overall the evaluation is designed to compare models using a consistent framework so that strengths and weaknesses can be clearly observed, by looking at multiple metrics together we ensure that the chosen model does not simply achieve high accuracy but instead demonstrates stable and balanced performance in distinguishing attrited customers from existing customers while remaining practical for decision support usage within the application.

Results and Discussion

As stated above we use the accuracy,precision, recall and F1 score with emphasis on ROC AUC when evaluating the performance of the models below is a table to show model performance comparison

Model	Accuracy	Precision	Recall	F1 score	ROC AUC
LR	0.8549	0.5311	0.8154	0.6432	0.9171
SVM	0.8781	0.5833	0.8400	0.6885	0.9349
RF	0.9501	0.9341	0.7415	0.8268	0.9852
NB	0.8682	0.5918	0.5754	0.5835	0.8351

The results show that the Random Forest model has the strongest overall performance with the highest accuracy and ROC AUC among all models, suggesting that ensemble learning is better at capturing nonlinear interactions and complex relationships within customer behavioural features, the high precision indicates that when Random Forest predicts a customer as attrited it is usually correct which protects against unnecessary interventions, however its recall is lower compared to Logistic Regression and SVM which shows that some churn cases are missed as the model is more conservative when assigning positive predictions.

The Support Vector Machine demonstrates strong discrimination capability as seen by its ROC AUC of 0.9349 and achieves the highest recall at 0.84 which means it successfully predicts a larger proportion of attrited customers compared to other models, this comes at the cost of lower

precision which increases false positives but may be acceptable in scenarios where missing potential churners is more costly than investigating additional customers. Logistic Regression provides stable baseline performance with ROC AUC of 0.9171 and relatively high recall, suggesting that linear relationships already capture meaningful churn patterns although its lower precision indicates limitations in modelling complex feature interactions compared to nonlinear approaches.

Gaussian Naive Bayes performs as the weakest model overall with the lowest ROC AUC and recall values, showing signs of reduced separation between classes and a higher tendency to miss attrited customers, this behaviour likely results from the independence assumptions underlying Naive Bayes which may not hold for financial behaviour features that are correlated. Comparison of ROC curves further supports these observations as Random Forest maintains the largest area under the curve followed by SVM and Logistic Regression while Naive Bayes lags behind.

Overall, Random Forest provides the strongest overall predictive performance and lowest false positive rate which is suitable for precise targeting, while SVM and Logistic Regression offer higher recall which may be more desirable in proactive churn prevention scenarios where detecting as many at-risk customers. Naive Bayes may be used as a supporting influence.

Conclusion

By connecting sophisticated data mining methods with a useful business interface, this project effectively created a thorough framework for forecasting and comprehending credit card customer attrition. We went beyond simple classification by employing both supervised learning and association rule mining to offer comprehensible insights into the particular behavioral patterns—like transaction frequency and utilization levels—that occur prior to a customer's departure.

Our analysis demonstrates that while models like SVM offer high sensitivity for determining the maximum number of at-risk individuals, ensemble approaches like Random Forest offer superior precision and overall separation capability. The project's main goal of converting analytical findings into a practical decision-support tool for banking professionals is achieved by integrating these models into a Streamlit application. In the end, banks can move from reactive observation to proactive intervention with this integrated approach, which may lower the high expenses related to customer turnover.

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Contribution

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